

Experiment 8 :

1. Setup Selenium in Jenkins

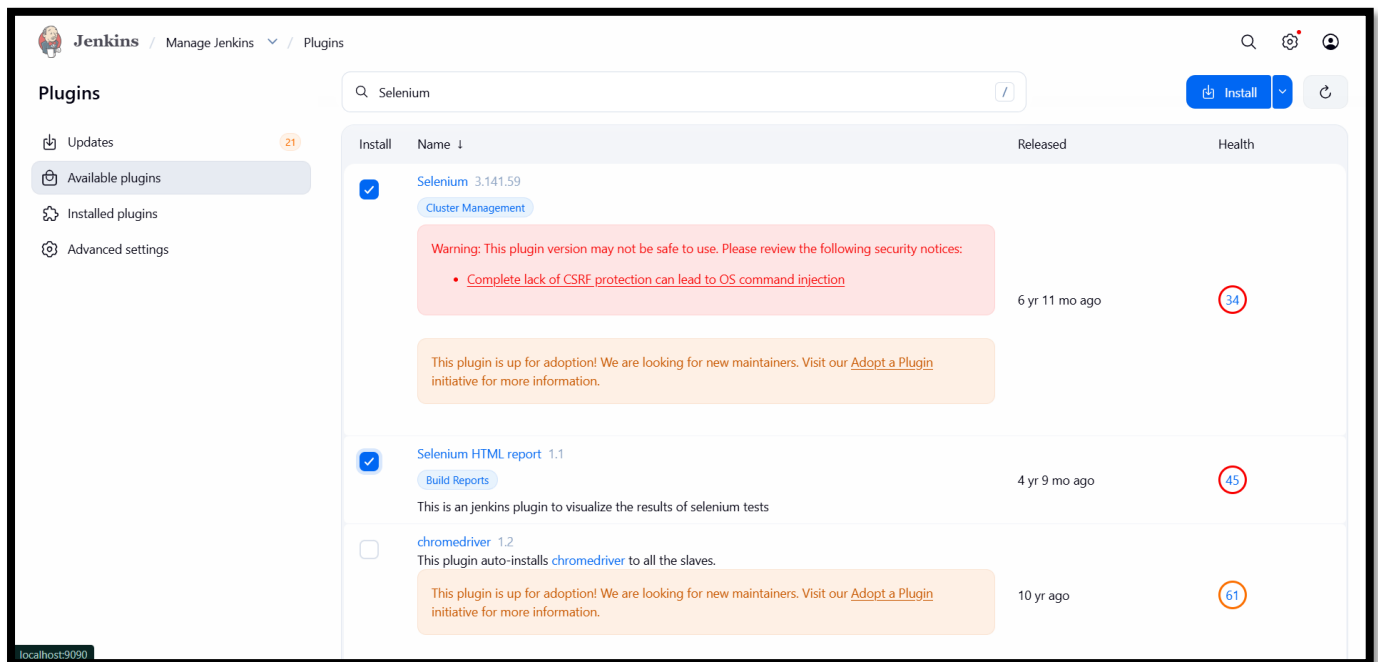
The screenshot shows the Jenkins dashboard. On the left, there is a sidebar with 'New Item', 'Build History', and a 'Build Queue' section. The 'Build Queue' section shows a table of builds. The table has columns for 'S' (Status), 'W' (Webhook), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The builds listed are 'hello_py', 'javaCalculator', 'javaFactorial', 'javaFibonacci', 'javaResume', 'Master-Slave', 'name_py', 'Pipeline1_Shivam', 'Pipeline2_Shivam', and 'Pipeline3_Shivam'. All builds are in a 'Success' state (green checkmark) and have a 'Last Success' time of 1 mo 15 days or less. The 'Last Duration' for each build is also shown.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	hello_py	1 mo 15 days #4	N/A	0.2 sec
✓	☀	javaCalculator	1 mo 6 days #1	N/A	1.6 sec
✓	☀	javaFactorial	1 mo 6 days #1	N/A	1.6 sec
✓	☀	javaFibonacci	1 mo 6 days #1	N/A	3 sec
✓	☀	javaResume	1 mo 5 days #3	N/A	1.5 sec
✓	☀	Master-Slave	29 days #3	N/A	0.31 sec
✓	☁	name_py	1 mo 15 days #2	1 mo 15 days #1	0.31 sec
✓	☀	Pipeline1_Shivam	21 days #1	N/A	3.8 sec
✓	☀	Pipeline2_Shivam	21 days #1	N/A	1 min 52 sec
✓	☀	Pipeline3_Shivam	21 days #1	N/A	10 sec

The screenshot shows the 'Manage Jenkins' page. At the top, there is a search bar and a 'Manage Jenkins' button. Below this, there are several sections: 'New version of Jenkins (2.557) is available for download (changelog).', 'Building on the built-in node can be a security issue. You should set the number of executors on the built-in node to 0. See the documentation.', 'Warnings have been published for the following currently installed components:', 'Jenkins 2.551 core and libraries: Multiple security vulnerabilities in Jenkins 2.554 and earlier, LTS 2.541.2 and earlier. A fix for this issue is available. Update Jenkins now.', 'Jenkins can enforce Content Security Policy (CSP). CSP tells web browsers what they are allowed to do while rendering a web page. This limits or even eliminates the impact of vulnerabilities like cross-site scripting (XSS). CSP is disabled by default for backward compatibility, but it is recommended to enable it, if possible.', and 'System Configuration' which includes 'System', 'Tools', and 'Plugins' sections.

System Configuration

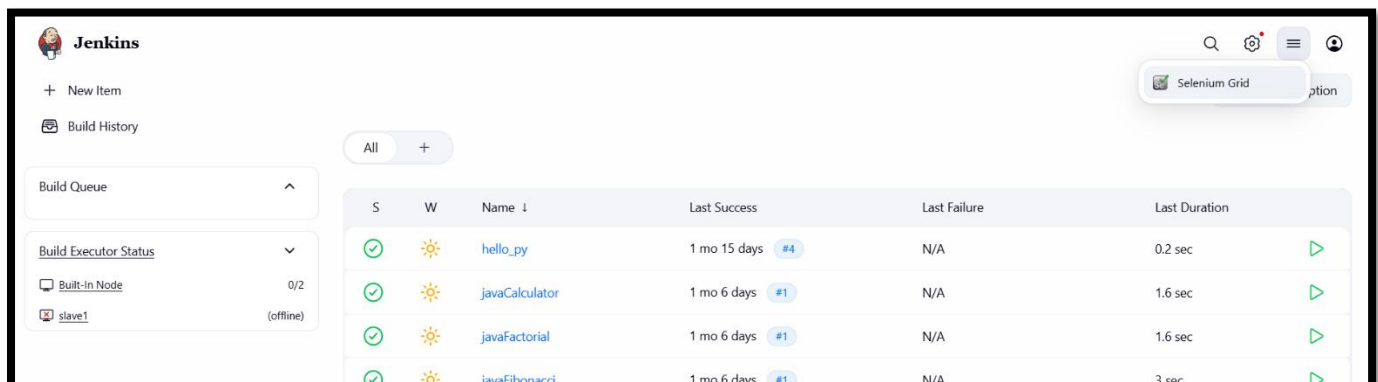
- System: Configure global settings and paths.
- Tools: Configure tools, their locations and automatic installers.
- Plugins: Add, remove, disable or enable plugins that can extend the functionality of Jenkins.



The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. A search bar at the top contains 'Selenium'. The left sidebar shows 'Available plugins' with a badge for 21 updates. The main table lists installed and available plugins. The 'Selenium' plugin (3.141.59) is highlighted with a red warning box: 'Warning: This plugin version may not be safe to use. Please review the following security notices: Complete lack of CSRF protection can lead to OS command injection'. It also has an orange adoption notice. The 'Selenium HTML report' (1.1) and 'chromedriver' (1.2) are also listed with adoption notices. Health status icons are shown in the 'Health' column.

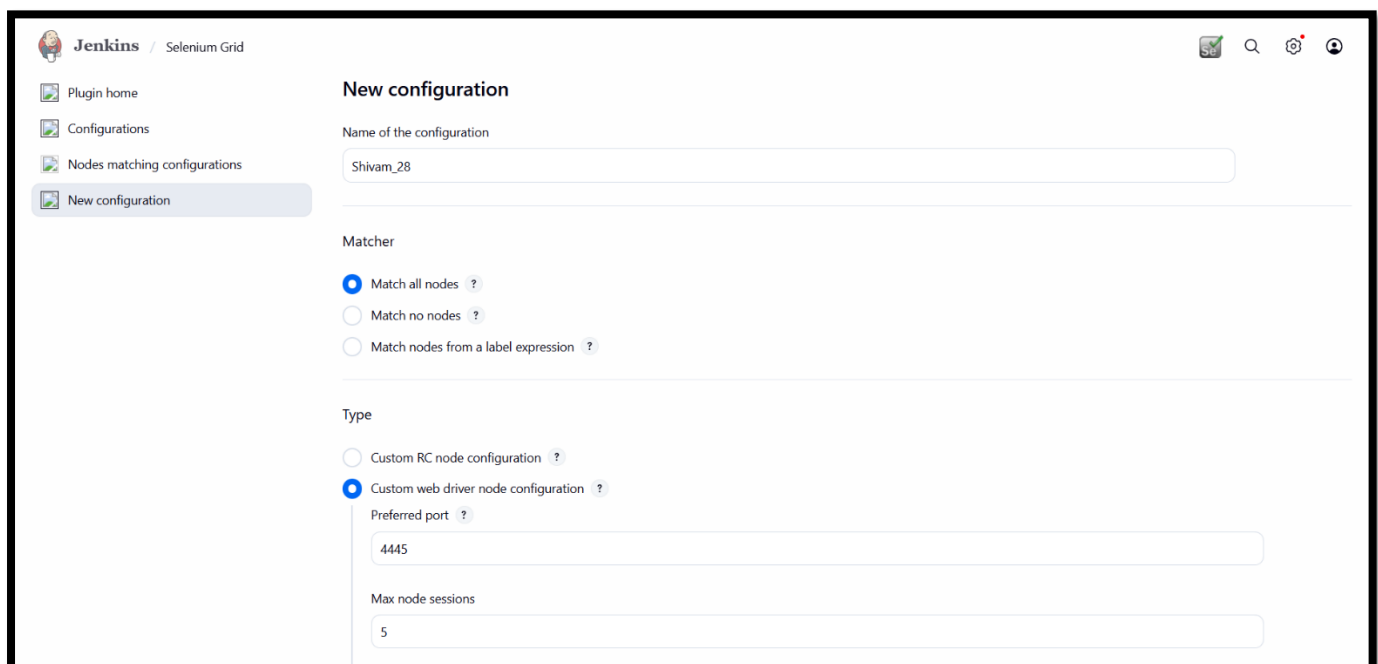
Install	Name	Released	Health
☒	Selenium 3.141.59 Cluster Management	6 yr 11 mo ago	34
☒	Selenium HTML report 1.1 Build Reports	4 yr 9 mo ago	45
☐	chromedriver 1.2	10 yr ago	61

2. Configure Selenium Grid



The screenshot shows the Jenkins 'Selenium Grid' view. The left sidebar shows 'Build Queue' and 'Build Executor Status' (0/2). The main table displays the status of various build executors.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	hello_py	1 mo 15 days #4	N/A	0.2 sec
✓	☀	javaCalculator	1 mo 6 days #1	N/A	1.6 sec
✓	☀	javaFactorial	1 mo 6 days #1	N/A	1.6 sec
✓	☀	javaFibonacci	1 mo 6 days #1	N/A	3 sec



The screenshot shows the 'New configuration' page for the Selenium Grid. The 'Name of the configuration' is 'Shivam_28'. The 'Matcher' section has 'Match all nodes' selected. The 'Type' section has 'Custom web driver node configuration' selected. The 'Preferred port' is '4445' and 'Max node sessions' is '5'.

Name of the configuration: Shivam_28

Matcher:

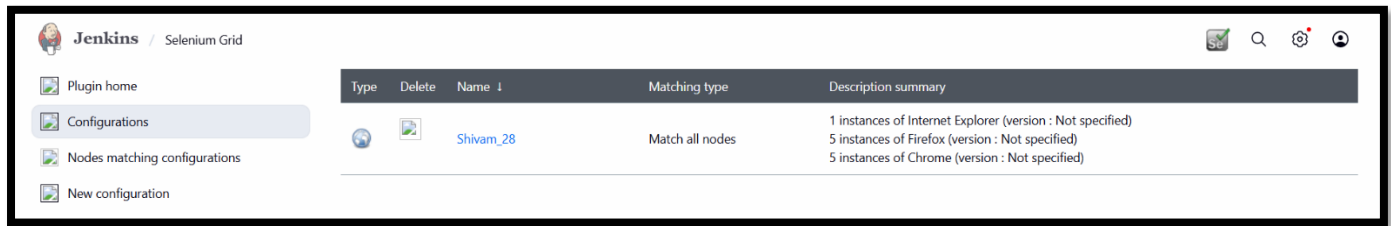
- ☒ Match all nodes ?
- ☐ Match no nodes ?
- ☐ Match nodes from a label expression ?

Type:

- ☐ Custom RC node configuration ?
- ☒ Custom web driver node configuration ?

Preferred port: 4445

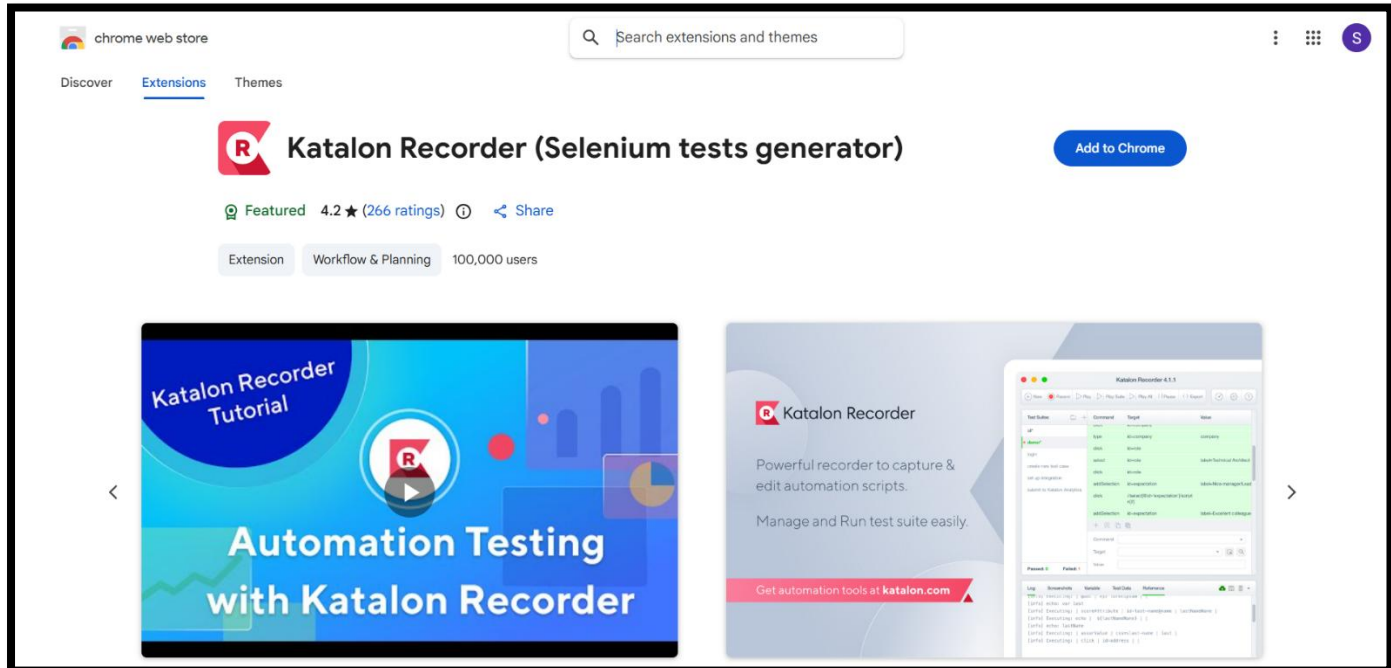
Max node sessions: 5



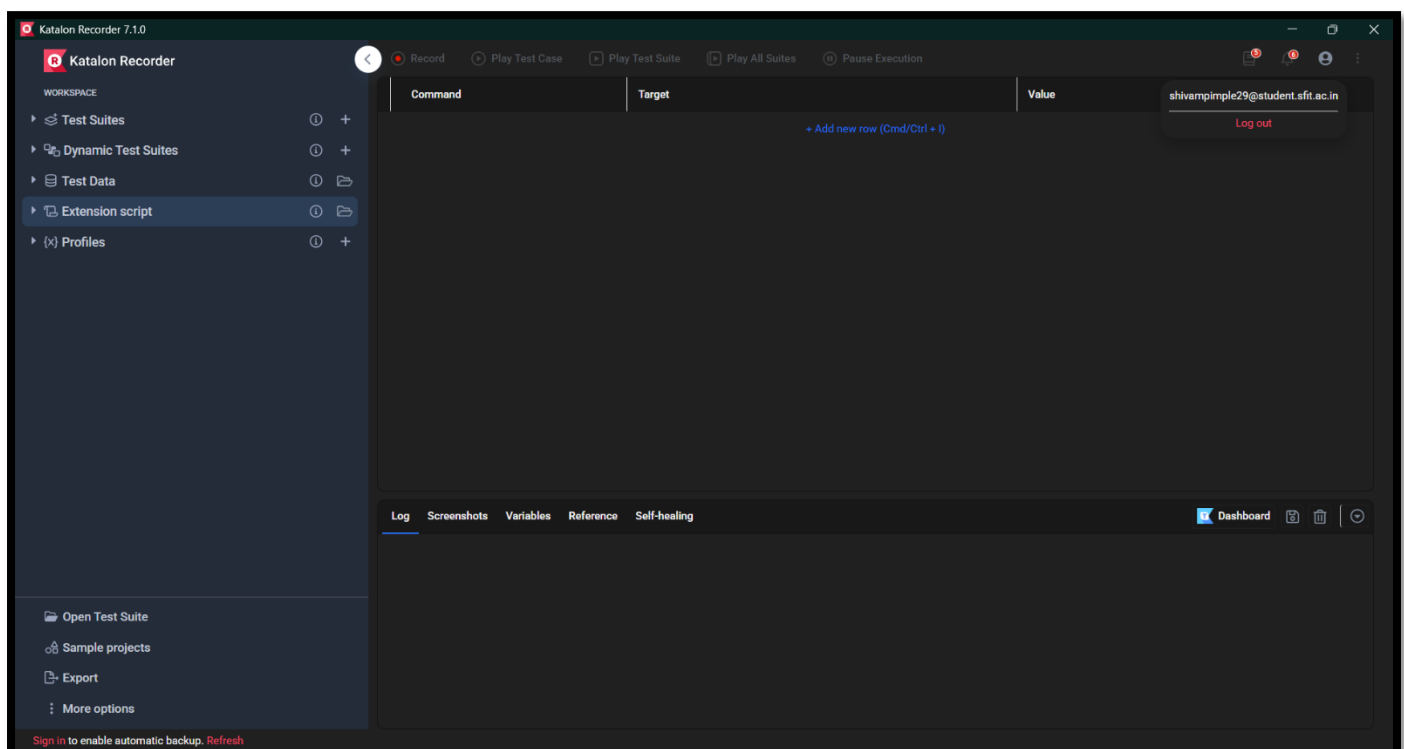
The screenshot shows the Jenkins Selenium Grid configuration page. On the left, there is a sidebar with links: Plugin home, Configurations (selected), Nodes matching configurations, and New configuration. The main area displays a table with columns: Type, Delete, Name, Matching type, and Description summary. A single configuration named 'Shivam_28' is listed with the matching type 'Match all nodes' and a description summary listing 1 instance of Internet Explorer, 5 instances of Firefox, and 5 instances of Chrome, all with version 'Not specified'.

Type	Delete	Name	Matching type	Description summary
		Shivam_28	Match all nodes	1 instances of Internet Explorer (version : Not specified) 5 instances of Firefox (version : Not specified) 5 instances of Chrome (version : Not specified)

3. Create test using Katalon

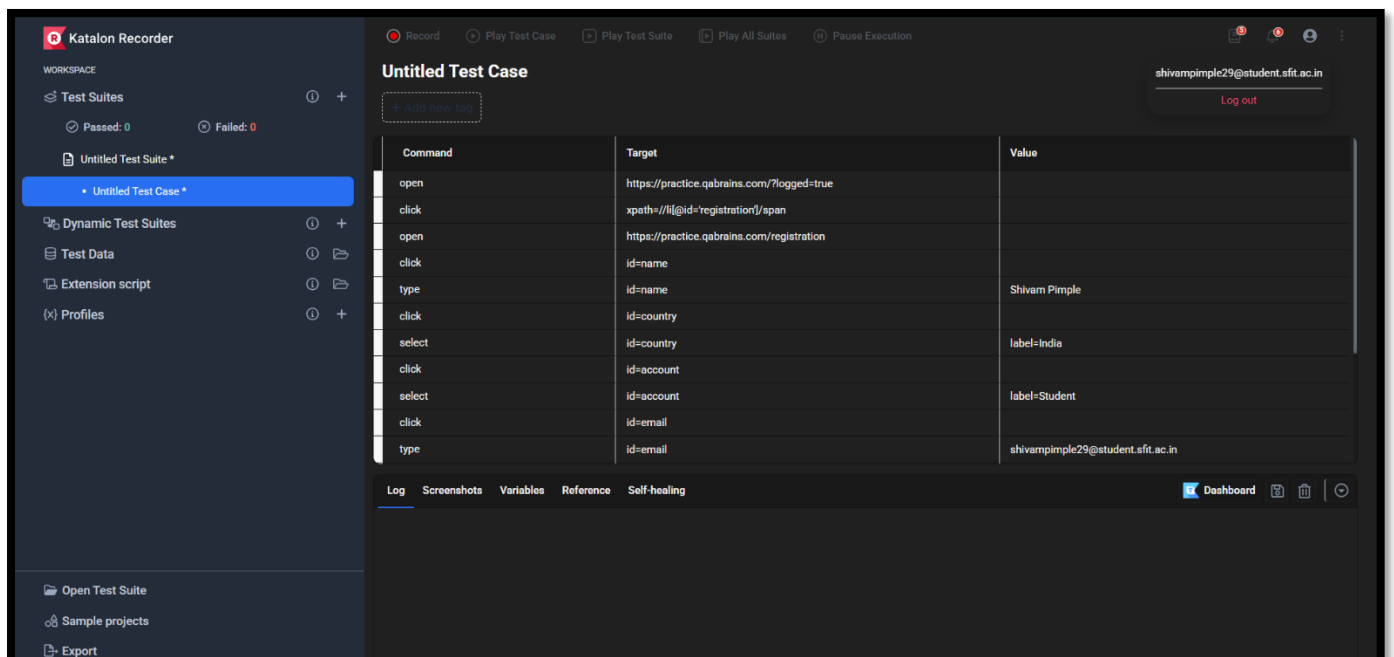
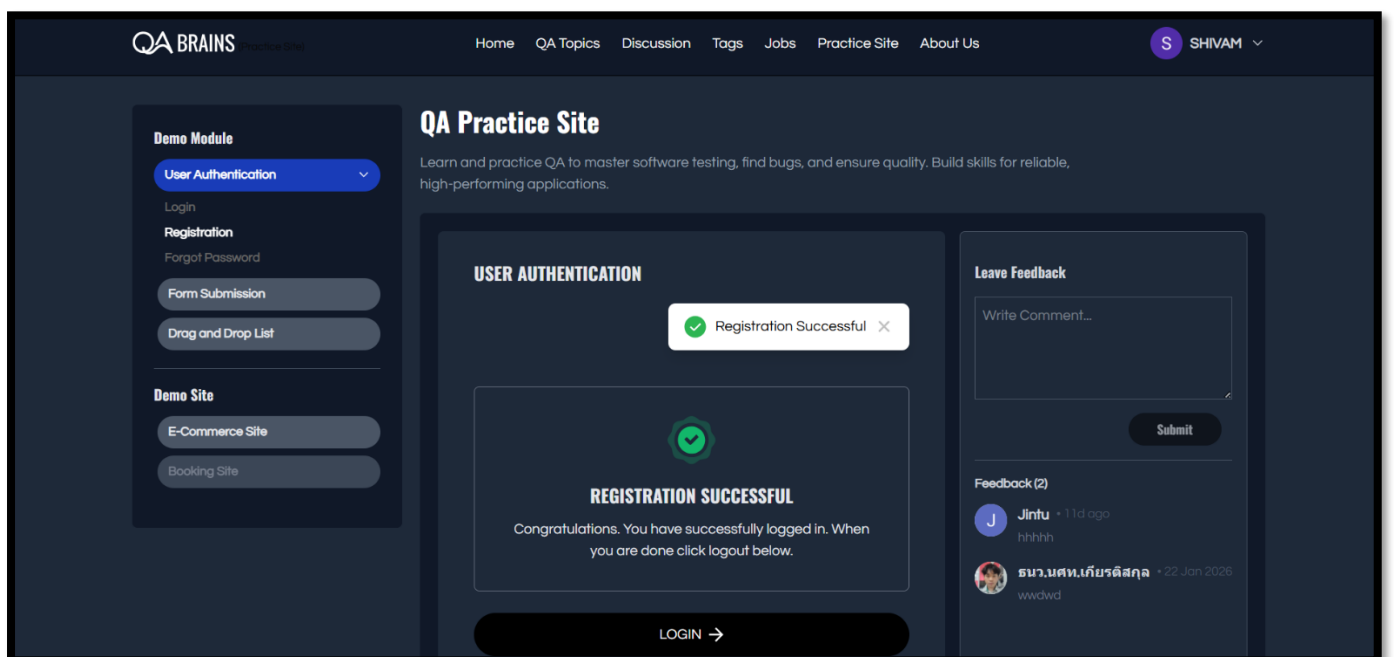
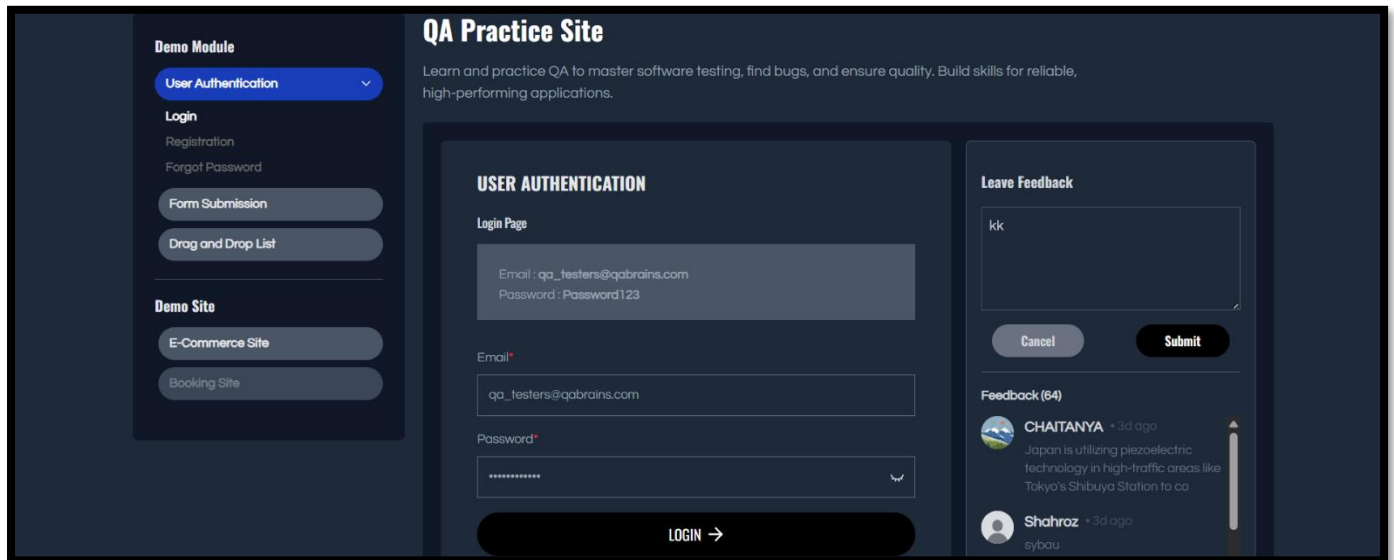


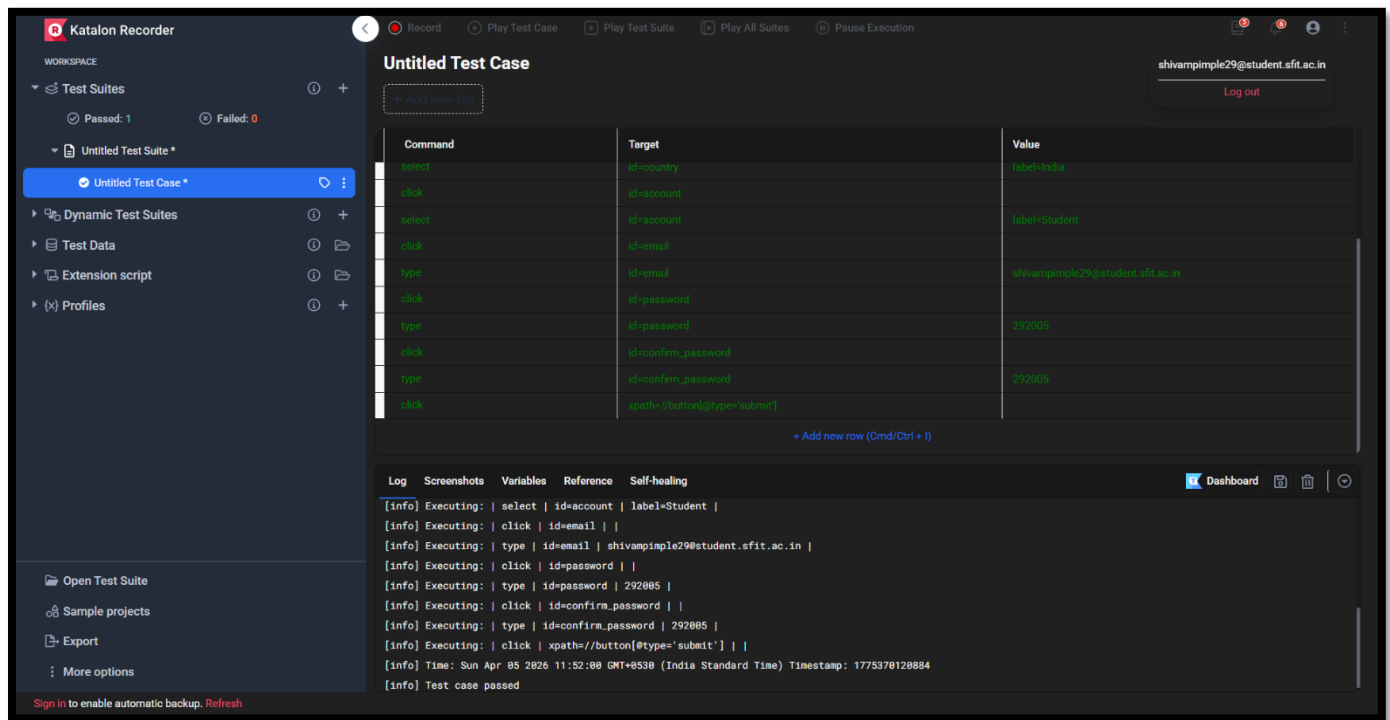
The screenshot shows the Katalon Recorder (Selenium tests generator) extension page on the Chrome Web Store. The page features the Katalon Recorder logo, a 'Featured' badge, a 4.2 star rating from 266 reviews, and a 'Share' button. Below this, there are three categories: Extension, Workflow & Planning, and 100,000 users. The main content area includes a large blue banner for 'Automation Testing with Katalon Recorder' and a smaller image showing the Katalon Recorder interface with a test suite list and a log.



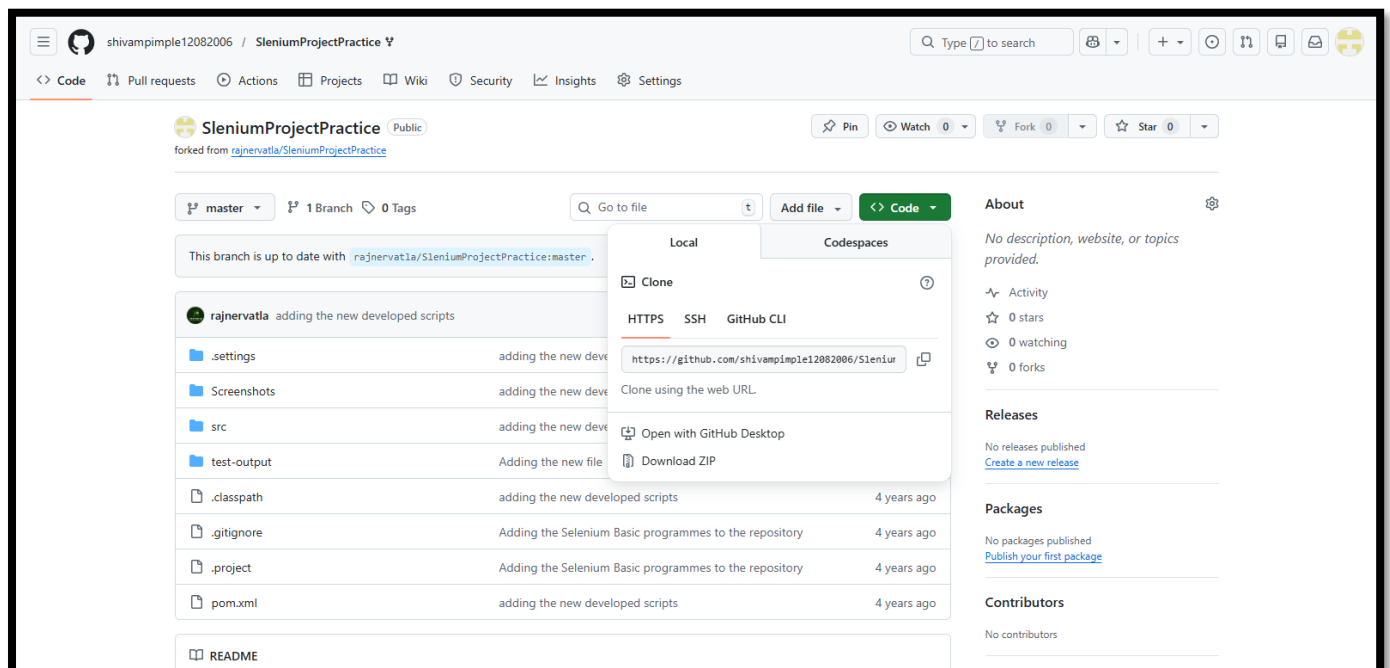
The screenshot shows the Katalon Recorder 7.1.0 interface. The left sidebar contains a 'WORKSPACE' section with links to Test Suites, Dynamic Test Suites, Test Data, Extension script (selected), and Profiles. Below this, there are links to Open Test Suite, Sample projects, Export, and More options. The main area displays a table with columns: Command, Target, and Value. The 'Value' column contains the text 'shivampimple29@student.sfi.ac.in' and a 'Log out' button. At the bottom, there is a 'Log' section with tabs for Log, Screenshots, Variables, Reference, and Self-healing. The 'Log' tab is active, showing a log of commands and their results.

Command	Target	Value
+ Add new row (Cmd/Ctrl + I)		

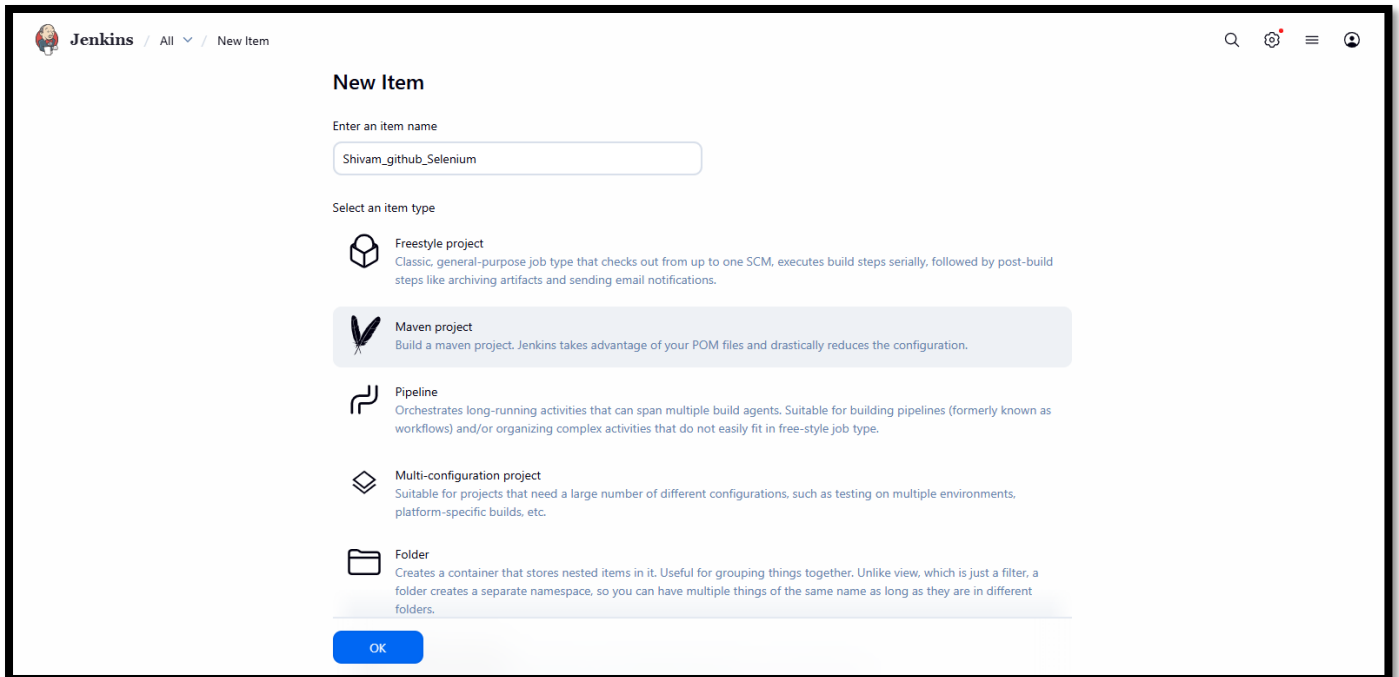




4. Push Test Project to GitHub



5. Setting Up and Executing a Jenkins Job with Post-Build Actions



The screenshot shows the Jenkins 'New Item' configuration page. At the top, the breadcrumb is 'Jenkins / All / New Item'. The main heading is 'New Item'. Below it, there is a text input field for 'Enter an item name' with the value 'Shivam_github_Selenium'. Underneath, a section titled 'Select an item type' lists five options: 'Freestyle project' (Classic, general-purpose job type), 'Maven project' (Build a maven project), 'Pipeline' (Orchestrates long-running activities), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations), and 'Folder' (Creates a container that stores nested items). The 'Maven project' option is highlighted with a light blue background. At the bottom, there is a blue 'OK' button.

New Item

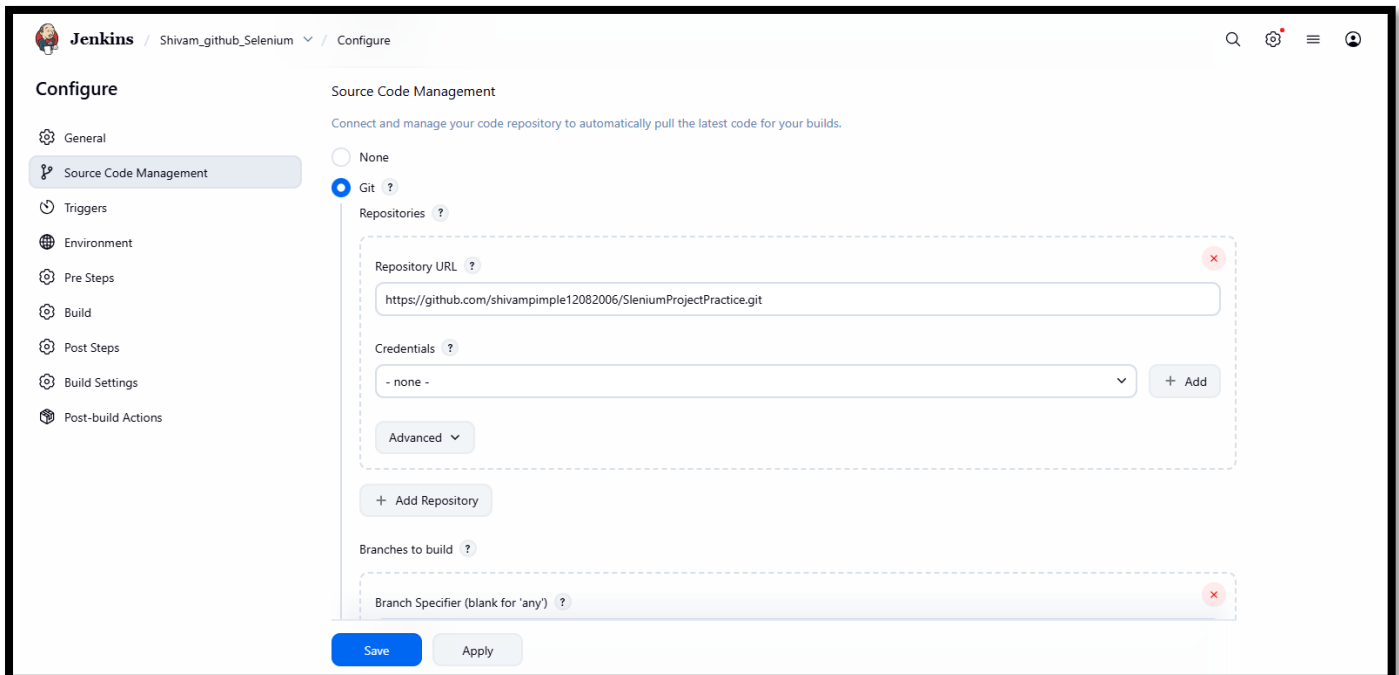
Enter an item name

Shivam_github_Selenium

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK



The screenshot shows the Jenkins 'Configure' page for the job 'Shivam_github_Selenium'. The breadcrumb is 'Jenkins / Shivam_github_Selenium / Configure'. The left sidebar shows the configuration tree with 'Source Code Management' selected. The main section is titled 'Source Code Management' and includes the instruction 'Connect and manage your code repository to automatically pull the latest code for your builds.' There are two radio buttons: 'None' and 'Git' (which is selected). Below the 'Git' button is a 'Repositories' section with a '+' icon and a question mark. It contains a 'Repository URL' field with the value 'https://github.com/shivampimple12082006/SeleniumProjectPractice.git', a 'Credentials' dropdown menu set to '- none -' with an 'Add' button, and an 'Advanced' dropdown menu. Below this is a '+ Add Repository' button. Further down is a 'Branches to build' section with a '+' icon and a question mark, containing a 'Branch Specifier (blank for 'any')' field. At the bottom, there are 'Save' and 'Apply' buttons.

Configure

General

Source Code Management

Triggers

Environment

Pre Steps

Build

Post Steps

Build Settings

Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/shivampimple12082006/SeleniumProjectPractice.git

Credentials ?

- none - + Add

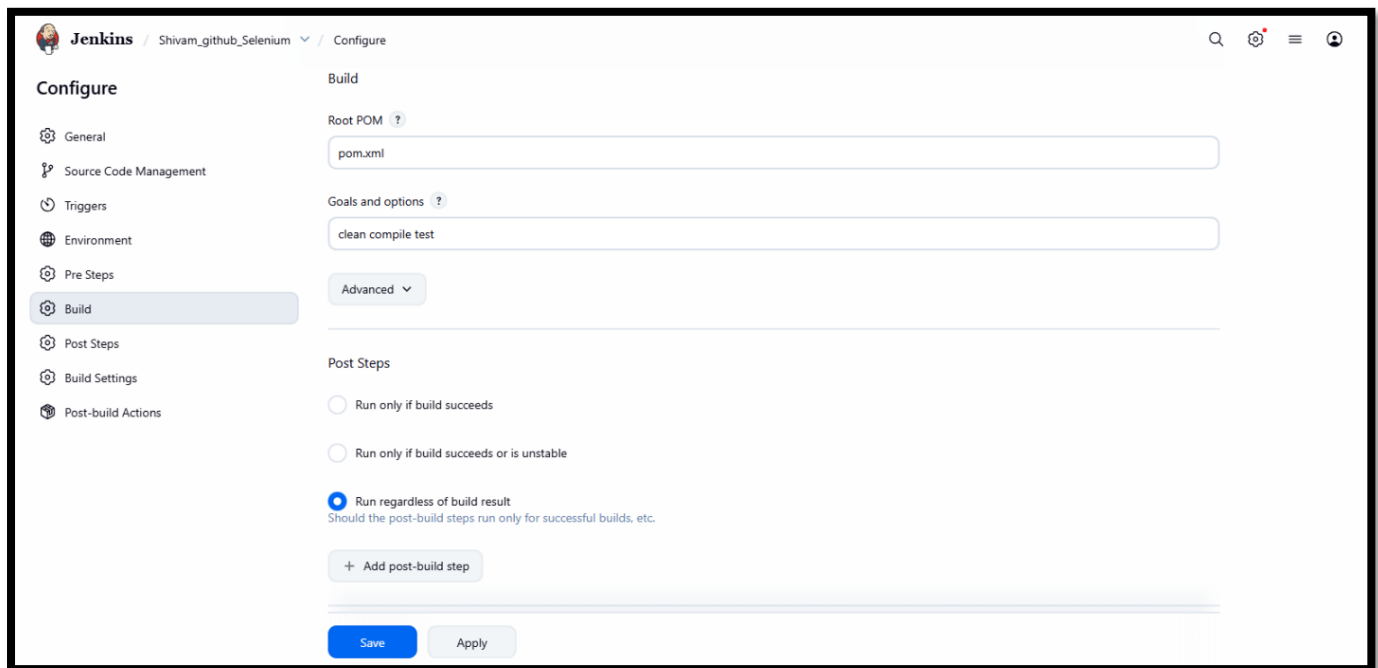
Advanced v

+ Add Repository

Branches to build ?

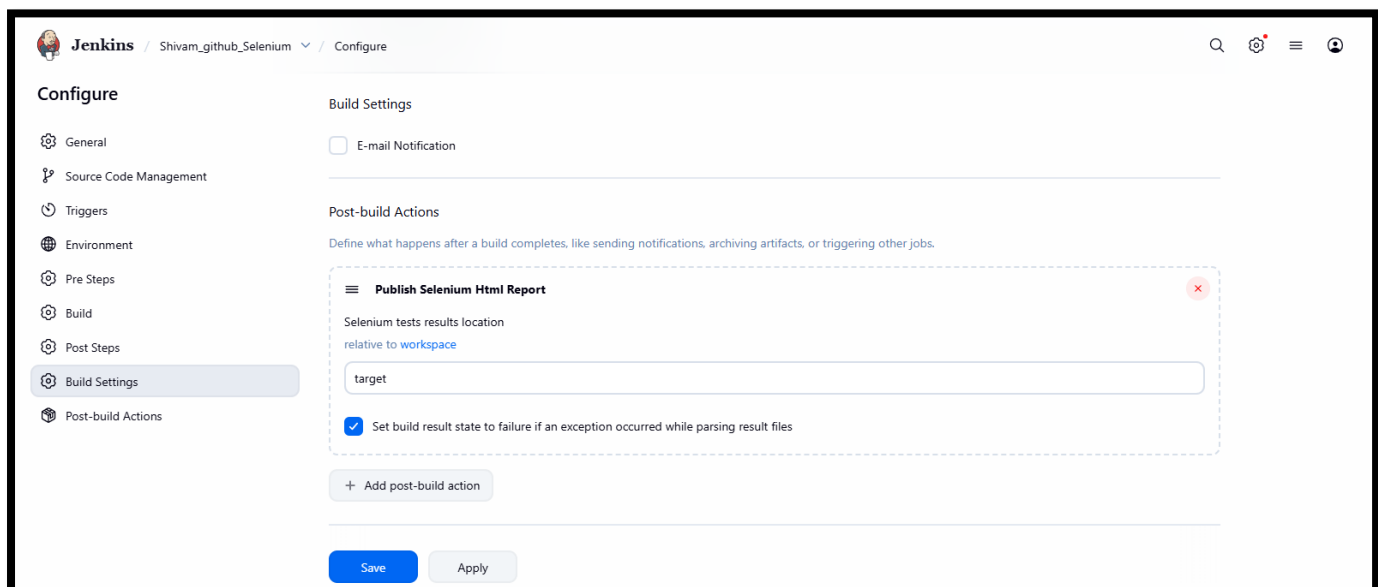
Branch Specifier (blank for 'any') ?

Save Apply



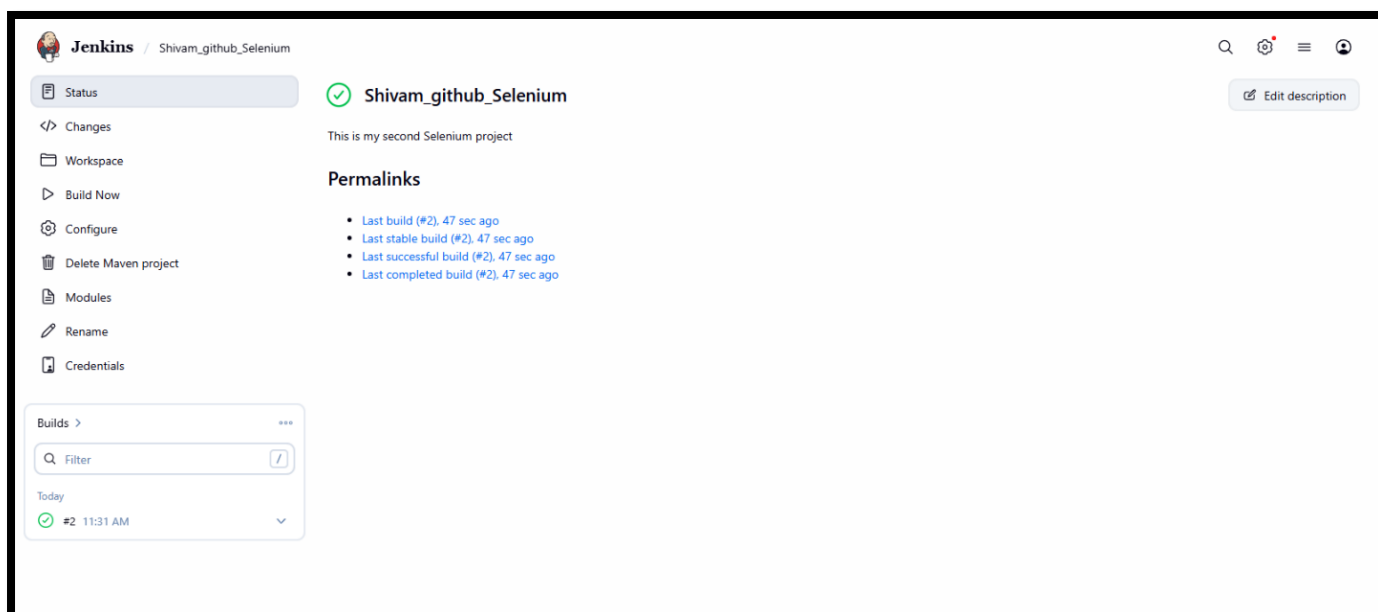
The screenshot shows the Jenkins configuration page for a job named 'Shivam_github_Selenium'. The 'Build' section is selected in the left sidebar. The configuration includes:

- Root POM:** pom.xml
- Goals and options:** clean compile test
- Advanced:** A dropdown menu.
- Post Steps:** Three radio buttons are present:
 - ☐ Run only if build succeeds
 - ☐ Run only if build succeeds or is unstable
 - ☒ Run regardless of build resultA note below states: "Should the post-build steps run only for successful builds, etc."
- Buttons:** '+ Add post-build step', 'Save', and 'Apply'.



The screenshot shows the Jenkins configuration page for 'Shivam_github_Selenium', with the 'Build Settings' and 'Post-build Actions' sections visible.

- Build Settings:** Includes an unchecked checkbox for 'E-mail Notification'.
- Post-build Actions:** A section titled 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' It contains a dashed box with the following configuration:
 - Publish Selenium Html Report:** A section with a red close button.
 - Selenium tests results location:** A text input field with the value 'target'.
 - Set build result state to failure if an exception occurred while parsing result files:** A checked checkbox.
- Buttons:** '+ Add post-build action', 'Save', and 'Apply'.



The screenshot shows the Jenkins job view page for 'Shivam_github_Selenium'. The 'Status' tab is selected in the left sidebar.

- Job Status:** A green checkmark icon and the text 'Shivam_github_Selenium'.
- Description:** 'This is my second Selenium project'.
- Permalinks:** A list of links for build history:
 - Last build (#2), 47 sec ago
 - Last stable build (#2), 47 sec ago
 - Last successful build (#2), 47 sec ago
 - Last completed build (#2), 47 sec ago
- Builds:** A section showing a list of builds. The first build is highlighted: '#2 11:31 AM'.

The screenshot shows the Jenkins web interface for build #2 of the 'Shivam_github_Selenium' job. The left sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build '#2', Timings, Git Build Data, Redeploy Artifacts, Selenium HTML Report, and See Fingerprints. The main area displays the console output, which includes the following text:

```
Started by user admin
Running as SYSTEM
Building on the built-in node in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Shivam_github_Selenium
The recommended git tool is: NONE
No credentials specified
> C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Shivam_github_Selenium\.git # timeout=10
Fetching changes from the remote Git repository
> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/shivampimple12082006/SeleniumProjectPractice.git # timeout=10
Fetching upstream changes from https://github.com/shivampimple12082006/SeleniumProjectPractice.git
> C:\Program Files\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.34.1.windows.1'
> C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/shivampimple12082006/SeleniumProjectPractice.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision e4d6ed8719a8f95934655f9a86feb825768cabf (refs/remotes/origin/master)
> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
> C:\Program Files\Git\bin\git.exe checkout -f e4d6ed8719a8f95934655f9a86feb825768cabf # timeout=10
Commit message: "adding the new developed scripts"
First time build. Skipping changelog.
Parsing POMs
Discovered a new module com.qa:Selenium Selenium
Modules changed, recalculating dependency graph
Established TCP socket on 59111
[Shivam_github_Selenium] $ "C:\Program Files\Java\jdk-17\bin\java" -cp C:\ProgramData\Jenkins\.jenkins\plugins\maven-plugin\WEB-INF\lib\maven35-agent-1.14.jar;C:\apache-maven-3.9.4-bin\apache-maven-3.9.4\boot\plexus-classworlds-2.7.0.jar;C:\apache-maven-3.9.4-bin\apache-maven-3.9.4\conf\logging-jenkins.maven3.agent.Maven35Main C:\apache-maven-3.9.4-bin\apache-maven-3.9.4 C:\ProgramData\Jenkins\war\WEB-INF\lib\remoting-3345.v03dee9b_f88fc.jar C:\ProgramData\Jenkins\.jenkins\plugins\maven-plugin\WEB-INF\lib\maven35-interceptor-1.14.jar C:\ProgramData\Jenkins\.jenkins\plugins\maven-
```

The screenshot shows the continuation of the Jenkins console output for build #2. The output includes the following text:

```
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ Selenium ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 9 source files with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[INFO] /C:/ProgramData/Jenkins/.jenkins/workspace/Shivam_github_Selenium/src/test/java/waits/FluentWaitExample.java: Some input files use or override a deprecated API.
[INFO] /C:/ProgramData/Jenkins/.jenkins/workspace/Shivam_github_Selenium/src/test/java/waits/FluentWaitExample.java: Recompile with -Xlint:deprecation for details.
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ Selenium ---
[JENKINS] Recording test results
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.215 s
[INFO] Finished at: 2026-03-30T11:32:13+05:30
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\Shivam_github_Selenium\pom.xml to com.qa:Selenium/0.0.1-SNAPSHOT/Selenium-0.0.1-SNAPSHOT.pom
channel stopped
Publishing Selenium report...
Copying the reports.
Finished: SUCCESS
```

Jenkins 2.533